

HIGH PERFORMANCE DATA PROCESSING

Week 4 Exercise

Designing High-Performance Data Analytics Architectures

1. Introduction

This exercise is designed to bridge the gap between theoretical concepts and real-world application in the domain of high-performance data processing. Students will examine how leading organisations and technology-driven systems manage, process, and derive value from large volumes of data. The exercise is aligned with the Week 4 topic on Big and Fast Data Analytics and the Role of High Performance Computing.

In today's data-driven environment, organisations must process enormous amounts of structured and unstructured data at high speeds to remain competitive. This requires a combination of cloud computing, edge computing, high performance computing (HPC), artificial intelligence (AI), and real-time processing pipelines. Through this exercise, students will develop a practical understanding of how these technologies are integrated into cohesive data analytics architectures.

Students will analyse a selected real-world case study, identify key data characteristics and challenges, and propose a complete system architecture that addresses the demands of modern high-performance data environments.

2. Learning Outcomes

Upon completing this exercise, students will be able to:

1. Identify and describe the data characteristics of a real-world high-performance data system, including volume, velocity, and variety.
2. Analyse key technical challenges associated with processing large-scale and real-time data in modern organisations.
3. Select and justify appropriate technologies for data ingestion, processing, storage, and analytics based on system requirements.
4. Design a complete and technically sound data analytics architecture that integrates cloud, edge, and high performance computing paradigms.
5. Communicate architectural decisions clearly, demonstrating an understanding of scalability, performance, and system design trade-offs.

3. Task Description

Students are required to work in pairs (two students per group). Each group must select one case study from the list provided and submit a comprehensive analysis along with a complete data analytics architecture design.

The selected case study should be analysed from a technical perspective, focusing on how the organisation or system processes large volumes of data efficiently and at scale. Students are expected to draw connections between the case study and the concepts taught in the course, particularly those relating to cloud computing, edge computing, HPC, AI, and streaming data pipelines.

Each group is required to produce a written report, an architecture diagram, and a brief oral presentation. The exercise encourages creative and well-reasoned design decisions, as there is no single universally correct architecture. Marks are awarded based on the quality of the analysis, the completeness and technical soundness of the architecture, and the clarity of justification.

Available Case Studies

Each group must select ONE case study from the following options:

No.	Case Study	Domain	Description
1	OpenAI	Artificial Intelligence	Large-scale AI model training using GPU clusters
2	Tesla	Autonomous Driving	Real-time vehicle sensor data processing
3	Netflix	Media Streaming	Recommendation system and user analytics
4	Grab	Mobility Services	Real-time ride matching and dynamic pricing
5	Shopee	E-Commerce	User behaviour and recommendation analytics
6	Google Maps	Navigation	Real-time traffic monitoring and route optimisation
7	Amazon	Cloud and E-Commerce	Large-scale cloud-based analytics and fulfilment
8	Meta Platforms	Social Media	Real-time user interaction and engagement analytics
9	Smart City System	Urban Technology	Traffic, IoT, and surveillance data analytics
10	Healthcare Monitoring System	Healthcare	Wearable devices and real-time health data processing

4. Detailed Instructions

The exercise is structured into four parts. Each part carries specific marks and must be addressed thoroughly.

Part A: Case Study Analysis (30 Marks)

A1. Data Characteristics (10 Marks)

Students must explain the nature of data within the selected case study. This includes:

- The types of data involved, such as batch, streaming, or real-time data.
- The sources of data, which may include IoT devices, users, sensors, transactions, or social media.
- The three dimensions of data: Volume (how much data is produced), Velocity (how fast data is generated), and Variety (the range of data formats and types).

A2. Key Challenges (10 Marks)

Students must identify and discuss at least three major technical challenges faced by the system. Examples include:

- Scalability: The ability of the system to grow and manage increased demand.
- Latency: The need to process and respond to data with minimal delay.
- Data processing speed: The requirement to handle high-velocity data efficiently.
- Storage limitations: Managing data retention, cost, and accessibility at scale.

A3. Technologies Required (10 Marks)

Students must identify and explain the core technologies that enable the system to operate. This should include:

- AI and Machine Learning tools used for prediction, recommendation, or classification.
- Streaming platforms, such as Apache Kafka or Apache Flink, if applicable.
- Storage systems, including data lakes, data warehouses, or data lakehouses.

Part B: Architecture Design (40 Marks)

This is the core component of the exercise. Students must produce a complete system architecture diagram that covers all six layers described in Section 5 of this document. The architecture must visually demonstrate the flow of data from source to output and must clearly incorporate cloud computing, edge computing (where applicable), and high performance computing. A hybrid architecture combining two or more computing paradigms is strongly encouraged.

Part C: Justification (20 Marks)

C1. Architecture Justification (10 Marks)

Students must explain why the proposed architecture is suitable for the selected case study. The justification should address why each component was chosen and how it contributes to meeting the system's requirements.

C2. Computing Paradigm Justification (10 Marks)

Students must explain the rationale behind the selection of computing paradigms used in the architecture. This should clarify why cloud computing, edge computing, HPC, or a hybrid combination was adopted and how each paradigm addresses specific performance or operational requirements.

Part D: Presentation (10 Marks)

Each group will deliver a short oral presentation of two to three minutes. The presentation should cover the key architectural decisions and demonstrate a clear understanding of the chosen design. Students are expected to explain the architecture confidently and concisely within the allocated time.

5. Architecture Requirements

The architecture design is the centrepiece of this exercise. To achieve full marks for Part B, students must ensure that the architecture diagram is complete, clearly labelled, and technically coherent. The following six layers must be present in the design.

Required Architecture Layers

6. **Identify all data inputs to the system. These may include sensors, user devices, application logs, IoT hardware, APIs, or external databases.**Data Sources:
7. **Describe how data is collected and brought into the system. This may involve batch ingestion pipelines or real-time streaming tools such as Apache Kafka.**Data Ingestion Layer:
8. **Explain how data is transformed and processed. Students should distinguish between batch processing, stream processing, and HPC-based parallel processing where applicable.**Processing Layer:
9. **Define the storage solution used, which may include object storage, distributed file systems, relational databases, data lakes, or data lakehouses.**Storage Layer:

10. **Describe how data is analysed and how AI or machine learning models are applied for tasks such as prediction, anomaly detection, or recommendation.** Analytics and AI Layer:
11. **Illustrate how results are delivered to end users or downstream systems, such as dashboards, APIs, prediction feeds, or alert systems.** Output Layer:

Important: The architecture must clearly indicate which components belong to cloud computing, edge computing, or high performance computing environments. A hybrid architecture that integrates two or more of these paradigms is highly encouraged and will be evaluated positively.

Architecture Tools

Students may use any of the following tools to create the architecture diagram:

- Draw.io (recommended for diagramming)
- Microsoft PowerPoint or Canva
- Any other diagram tool of the group's choice

Hand-drawn diagrams are also acceptable, provided they are neat, clearly labelled, and professionally presented.

Bonus Component (Optional, up to 5 Additional Marks)

Students who wish to earn additional marks may include any of the following in their architecture:

- A complete AI model pipeline showing both the training and inference stages.
- A data lakehouse design that incorporates features of both a data lake and a data warehouse.
- A real-time dashboard output layer demonstrating how processed results are visualised.

6. Deliverables

Each group is required to submit the following three components:

Submission Requirements

12. **A concise written report addressing all four parts of the exercise. The report must be clearly structured, professionally written, and submitted in either PDF or Word format.** Written Explanation (1 to 2 pages):
13. **A clearly labelled system architecture diagram showing all required layers and computing paradigms. This must be included within the written report or submitted as a separate file.** Architecture Diagram (Mandatory):
14. **A brief summary, which may be written in paragraph or structured point form, explaining the rationale behind key design decisions. This may be incorporated into the written report or presented as a separate section.** Justification Summary:

All submissions must be made by the designated deadline through the course submission platform. Late submissions may be subject to a mark deduction as per the course policy.

7. Evaluation Criteria

The exercise will be evaluated using the rubric below. Marks are distributed across four components, totalling 100 marks.

Part	Component	Criteria	Marks
A	Case Study Analysis	Depth, accuracy, and relevance of analysis	30
B	Architecture Design	Completeness, clarity, and technical soundness	40
C	Justification	Logical reasoning and alignment to requirements	20
D	Presentation	Clarity, confidence, and adherence to time	10
	TOTAL		100

Each component will be assessed based on the following general criteria:

- Analysis: The depth, accuracy, and relevance of the case study investigation.
- Architecture: The technical completeness, correctness, and clarity of the proposed design.
- Justification: The logical reasoning provided for technology and paradigm selections.
- Presentation: The clarity, confidence, and time management demonstrated during the oral presentation.

8. Student Guidance

The following guidance is provided to help students approach the exercise effectively and maximise their performance.

Focus Areas

- Prioritise real-time versus batch processing distinctions, as these are central to the case studies.
- Consider scalability and performance throughout the design process.
- Integrate AI and HPC components where they are relevant to the case study.
- Ensure that your architecture reflects the specific demands of the selected system, rather than applying a generic template.

Justification Approach

- Every design decision should be accompanied by a clear rationale.
- Explain why a particular technology was chosen over alternatives.
- Relate your choices back to the data characteristics and challenges identified in Part A.

Design Philosophy

- There is no single correct architecture for any of the case studies.
- Marks are awarded for logical reasoning, technical depth, and clarity of communication.
- Original and well-reasoned designs will be viewed favourably, even if they differ from conventional approaches.
- Strive for a design that is both technically sound and practically feasible.

This exercise directly supports the Week 4 learning outcomes on the evolution of data architectures from 2000 to 2025, the comparison of cloud, edge, and HPC paradigms, and the connection between industry case studies and architecture design. Students are encouraged to revisit lecture materials and relevant readings before beginning the analysis.

This exercise helps you bridge theory and real-world systems by designing how modern platforms use high-performance data analytics to process massive data efficiently.